Zirui Zheng

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EDUCATION

New York University Tandon School of Engineering — Brooklyn, New York, USA
Master of Science in Computer Engineering — Sep 2025 - Jun 2027

- GPA: 4.0 / 4.0
- **Relevant Coursework:** Machine Learning, Real Time Embedded Systems, Computing Systems Architecture, Parallel and Custom Computer Architecture, Introduction to Systems Engineering

Hunter College, CUNY — New York, NY
Bachelor of Arts in Computer Science, Minor in Mathematics — Jun 2021 - Jun 2025

- GPA: 3.68 / 4.0
- Dean's Honor 2021 - 2024
- **Relevant Coursework:** C++ Programming, Database Management, Big Data, SQL, BlockChain

SKILLS

Certifications: AWS Solution Architect Associate, AWS - July 2024 (Certified Link)
Programming Language: C++, Python, SQL, JavaScript, Terraform, GitHub Actions (YAML scripting), HTML, CSS, Bash, GDScript
Technologies & Tools: Git/GitHub, AWS, Azure, GCP, CI/CD, Infrastructure as Code (Terraform), Kubernetes, Docker, MongoDB, Cypress, CloudFlare, Linux, Spyder, Cloud Computing, Web Services
Embedded & ML: Mbed OS, ARM CMSIS-DSP, BLE, I2C, scikit-learn, PyTorch, NumPy, Pandas
Spoken Languages: Mandarin (Native)

PROJECTS

Parkinson's Disease Tremor Detection System, Link | C++, Mbed OS, ARM CMSIS-DSP, BLE, I2C
Sep 2025 – Dec 2025

- Developed an embedded system on a development board with LSM6DSL IMU to detect Parkinson's disease symptoms (tremor, dyskinesia, freezing of gait) using real-time accelerometer/gyroscope data.
- Implemented FFT-based frequency analysis (52 Hz sampling, 128-point FFT) to classify symptoms: tremor (3–5 Hz), dyskinesia (5–7 Hz), and freezing gait via step detection.
- Integrated BLE communication to transmit detection results via three separate characteristics (tremor, dyskinesia, FOG) to a mobile device, and used onboard LEDs for visual feedback.
- Designed robust data processing pipeline with interrupt-driven sensor reads and timer-based freezing detection, ensuring repeatability and low-latency response.

32-bit RISC-V Processor Simulator (Single & Five-Stage Pipeline) | Python, Computer Architecture
Sep 2025 – Dec 2025

- Built a cycle-accurate interpreter for the RISC-V RV32I instruction set, supporting both single-cycle and five-stage pipelined implementations.
- Implemented instruction fetch, decode, execution, memory access, and writeback stages; handled data hazards via forwarding and control hazards via branch prediction/flushing.
- Extended professor-provided boilerplate code to fully decode all R/I/B/S/U/J instruction types, execute ALU operations, and manage register file/memory state.
- Output performance metrics (cycles, instructions, CPI, IPC) and per-cycle pipeline state for validation against test cases; achieved correct execution for all provided test programs.

Zirui-Resume-website, Portfolio | Python, JavaScript, AWS, Terraform, CI/CD — Feb 2024 - June 2024

- Personal website leveraging AWS infrastructure with a real-time visitor counter.
- Developed the user interface using HTML, CSS, and JavaScript; hosted files using S3 and distributed via CloudFront.
- Developed Python scripts for AWS Lambda to retrieve data from DynamoDB and create a visitor counter.
- Implemented Terraform and automated deployment via GitHub CI/CD pipeline for AWS resources.
- Employed Cypress for testing updated code before deployment to AWS.

PROFESSIONAL EXPERIENCE

***Participant*, Google Software Engineering Program (G-SWEP)** New York, NY
Basta & Google Oct 2024 – Dec 2024

- Selected participants over thousands of applicants to pair program with a Google Software Engineer for 10 weeks to receive 1-to-1 mentorship to prepare for technical and behavioral interviews.
- Developed expertise with industry knowledge on system design, algorithm analysis, unit testing, and debugging skills.
- Immersed myself in Google's engineering culture and learned best practices for software development

Python Teaching Assistant | Python Libraries (Panda, Matplotlib, Numpy, Folium) New York, NY
CUNY Hunter College Sep 2023 – May 2024

- Facilitated lecture sessions for over 100 students, enhancing understanding of Python and its libraries, resulting in improved comprehension and application of Python concepts.
- Guided students through complex algorithmic problem-solving, fostering enhanced problem-solving and debugging skills within the tech domain.
- Articulated complex concepts engagingly and clearly to diverse groups, utilizing various methods, including visual aids and interactive discussions, to ensure students understand challenging material effectively.